Reporte Oficial de Mantenibilidad y Deuda Técnica

Análisis Estático y Financiero del Código Fuente (Arquitectura Limpia, SOLID, PHAME) — Actualizado: 25/10/2026

DEUDA TÉCNICA TOTAL (2)

10,052.30 hs

Equivalente a **62.83 meses** (4) sobre 105 archivos (16,392 LOC)

COSTO TOTAL DE REPARACIÓN (3)

$ 201,045.60

Calculado a razón de $20 USD / hora de desarrollo

ÍNDICE MANTENIBILIDAD PROMEDIO (1)

33.53 / 100

Interés Anual Base (5): **$1,400 USD** por módulo

VARIABLES GLOBALES (GLOSARIO)

- **LOC:** Líneas de código (sin blancos ni comentarios)
- **G:** Complejidad ciclomática de McCabe
- **K (Factor de Corrección):** 0.01
- **Salario por Hora:** $20.00 USD
- **Horas por Mes:** 160 horas laborables

✅ Plantilla Oficial de 7 Fórmulas Aplicada

```
1. MI = max(0, ((171 - 25·ln(LOC) - 0.23·G)/171)·100)
2. Deuda (Hs) = max(0, (75 - MI)·LOC·K)
3. Costo = Deuda (Hs) × Salario/Hora
4. Meses = Deuda (Hs) / Horas_Mes
5. Interés Anual = 10 × (14 - 7) × Salario
6. Payback = Costo / Interés Anual
7. ROI = 100 × (((Interés Anual × 2) - Costo) / Costo)
```

Archivos con Mayor Deuda Técnica (Top 8 Prioritarios)

Código Fuente Propio (Sin venv / node_modules)

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
`js/accounts.ts`	1268	462	**0.00**	951.00 hs	$19,020.00	5.94 m	$1,400	13.59 a	**-85.3%**
`js/shopping.ts`	911	265	**0.00**	683.25 hs	$13,665.00	4.27 m	$1,400	9.76 a	**-79.5%**
`js/services/scannerService.ts`	827	312	**0.00**	620.25 hs	$12,405.00	3.88 m	$1,400	8.86 a	**-77.4%**
`js/goals.ts`	755	200	**0.00**	566.25 hs	$11,325.00	3.54 m	$1,400	8.09 a	**-75.3%**
`js/budgets.ts`	695	173	**0.00**	521.25 hs	$10,425.00	3.26 m	$1,400	7.45 a	**-73.1%**
`js/supermercados.ts`	675	170	**0.00**	506.25 hs	$10,125.00	3.16 m	$1,400	7.23 a	**-72.3%**
`js/reports.ts`	468	74	**0.16**	350.27 hs	$7,005.31	2.19 m	$1,400	5.00 a	**-60.0%**
`js/transactions.ts`	413	170	**0.00**	309.75 hs	$6,195.00	1.94 m	$1,400	4.42 a	**-54.8%**

** Nota: Este reporte audita exclusivamente los archivos fuente propios del repositorio (directorios api/ y js/), excluyendo librerías de terceros (venv, fluxo_venv, node_modules) para reflejar el costo operativo real.*

Métricas Detalladas por Archivo (1/4)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
api/wallet_adapters/mercadopago_adapter.py	360	65	**5.20**	251.27 hs	$5,025.37	1.57 m	$1,400	3.59 a	**-44.3%**
api/services/sepa/comparador.py	303	45	**10.41**	195.70 hs	$3,913.96	1.22 m	$1,400	2.80 a	**-28.5%**
api/services/recommendation.py	292	63	**8.53**	194.08 hs	$3,881.68	1.21 m	$1,400	2.77 a	**-27.9%**
api/routers/wallets.py	265	46	**12.24**	166.32 hs	$3,326.39	1.04 m	$1,400	2.38 a	**-15.8%**
api/routers/accounts.py	235	36	**15.34**	140.20 hs	$2,804.05	0.88 m	$1,400	2.00 a	**-0.1%**
api/tests/test_sepa.py	240	1	**19.74**	132.63 hs	$2,652.52	0.83 m	$1,400	1.89 a	**5.6%**
api/services/auth_service.py	208	46	**15.78**	123.18 hs	$2,463.61	0.77 m	$1,400	1.76 a	**13.7%**
api/services/shopping/generic_extractor.py	184	84	**12.46**	115.07 hs	$2,301.47	0.72 m	$1,400	1.64 a	**21.7%**
api/services/shopping_service.py	191	51	**16.35**	112.02 hs	$2,240.33	0.70 m	$1,400	1.60 a	**25.0%**
api/services/ai_service.py	185	44	**17.76**	105.89 hs	$2,117.85	0.66 m	$1,400	1.51 a	**32.2%**
api/routers/sepa.py	194	9	**21.77**	103.26 hs	$2,065.17	0.65 m	$1,400	1.48 a	**35.6%**
api/models.py	187	1	**23.39**	96.52 hs	$1,930.31	0.60 m	$1,400	1.38 a	**45.1%**
api/services/shopping/mercadolibre_extractor.py	162	75	**15.53**	96.34 hs	$1,926.76	0.60 m	$1,400	1.38 a	**45.3%**
api/infrastructure/repositories/sqlalchemy_pri…	178	24	**21.01**	96.09 hs	$1,921.87	0.60 m	$1,400	1.37 a	**45.7%**
api/seed_dummy.py	180	13	**22.33**	94.80 hs	$1,896.08	0.59 m	$1,400	1.35 a	**47.7%**
api/main.py	176	17	**22.12**	93.07 hs	$1,861.32	0.58 m	$1,400	1.33 a	**50.4%**
api/alembic/versions/4faf7655fc78_baseline_mig…	181	1	**23.86**	92.56 hs	$1,851.12	0.58 m	$1,400	1.32 a	**51.3%**
api/routers/auth.py	171	16	**22.68**	89.47 hs	$1,789.43	0.56 m	$1,400	1.28 a	**56.5%**
api/services/sepa/loader.py	166	30	**21.23**	89.26 hs	$1,785.22	0.56 m	$1,400	1.28 a	**56.8%**
api/services/account_service.py	163	36	**20.69**	88.53 hs	$1,770.58	0.55 m	$1,400	1.26 a	**58.1%**
api/services/goal_service.py	158	47	**19.66**	87.43 hs	$1,748.62	0.55 m	$1,400	1.25 a	**60.1%**
api/services/transaction_service.py	126	27	**25.66**	62.17 hs	$1,243.30	0.39 m	$1,400	0.89 a	**125.2%**
api/tests/test_accounts.py	131	5	**28.05**	61.50 hs	$1,230.02	0.38 m	$1,400	0.88 a	**127.6%**
api/infrastructure/migrations.py	121	16	**27.73**	57.19 hs	$1,143.84	0.36 m	$1,400	0.82 a	**144.8%**
api/schemas.py	117	1	**30.24**	52.37 hs	$1,047.31	0.33 m	$1,400	0.75 a	**167.4%**
api/tests/test_goals.py	110	1	**31.15**	48.24 hs	$964.81	0.30 m	$1,400	0.69 a	**190.2%**
api/services/sepa/promociones.py	95	11	**31.94**	40.90 hs	$818.08	0.26 m	$1,400	0.58 a	**242.3%**
api/services/sepa/downloader.py	81	9	**34.54**	32.77 hs	$655.40	0.20 m	$1,400	0.47 a	**327.2%**

Métricas Detalladas por Archivo (2/4)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
api/services/sepa/ingestion.py	79	4	**35.58**	31.14 hs	$622.82	0.19 m	$1,400	0.44 a	**349.6%**
api/application/use_cases/sepa_use_cases.py	74	11	**35.60**	29.16 hs	$583.19	0.18 m	$1,400	0.42 a	**380.1%**
api/security.py	73	15	**35.26**	29.01 hs	$580.26	0.18 m	$1,400	0.41 a	**382.5%**
api/tests/test_recommendation.py	65	13	**37.22**	24.56 hs	$491.11	0.15 m	$1,400	0.35 a	**470.1%**
api/tests/test_budgets.py	64	3	**38.79**	23.17 hs	$463.44	0.14 m	$1,400	0.33 a	**504.2%**
api/database.py	60	18	**37.72**	22.37 hs	$447.36	0.14 m	$1,400	0.32 a	**525.9%**
api/infrastructure/dependencies.py	60	1	**40.01**	21.00 hs	$419.92	0.13 m	$1,400	0.30 a	**566.8%**
api/wallet_adapters/plaid_adapter.py	57	2	**40.62**	19.60 hs	$391.91	0.12 m	$1,400	0.28 a	**614.5%**
api/routers/ai.py	56	8	**40.07**	19.56 hs	$391.17	0.12 m	$1,400	0.28 a	**615.8%**
api/domain/entities/product.py	57	1	**40.76**	19.52 hs	$390.38	0.12 m	$1,400	0.28 a	**617.3%**
api/wallet_adapters/belvo_adapter.py	56	2	**40.88**	19.11 hs	$382.13	0.12 m	$1,400	0.27 a	**632.7%**
api/infrastructure/repositories/sqlalchemy_use…	52	7	**41.29**	17.53 hs	$350.57	0.11 m	$1,400	0.25 a	**698.7%**
api/routers/shopping.py	52	1	**42.10**	17.11 hs	$342.17	0.11 m	$1,400	0.24 a	**718.3%**
api/wallet_adapters/base_adapter.py	49	2	**42.83**	15.76 hs	$315.24	0.10 m	$1,400	0.23 a	**788.2%**
api/routers/goals.py	47	3	**43.31**	14.90 hs	$297.91	0.09 m	$1,400	0.21 a	**839.9%**
api/services/sepa/geo.py	44	6	**43.87**	13.70 hs	$273.96	0.09 m	$1,400	0.20 a	**922.1%**
api/infrastructure/repositories/sqlalchemy_wal…	44	1	**44.54**	13.40 hs	$268.04	0.08 m	$1,400	0.19 a	**944.6%**
api/routers/transactions.py	42	1	**45.22**	12.51 hs	$250.14	0.08 m	$1,400	0.18 a	**1019.4%**
api/alembic/env.py	40	12	**44.45**	12.22 hs	$244.36	0.08 m	$1,400	0.17 a	**1045.8%**
api/services/sepa/cleaner.py	40	11	**44.59**	12.16 hs	$243.28	0.08 m	$1,400	0.17 a	**1050.9%**
api/services/ocr.py	40	5	**45.40**	11.84 hs	$236.83	0.07 m	$1,400	0.17 a	**1082.3%**
api/services/budget_service.py	39	7	**45.50**	11.51 hs	$230.12	0.07 m	$1,400	0.16 a	**1116.8%**
api/infrastructure/repositories/sqlalchemy_tra…	38	1	**46.68**	10.76 hs	$215.20	0.07 m	$1,400	0.15 a	**1201.1%**
api/infrastructure/repositories/sqlalchemy_goa…	36	1	**47.47**	9.91 hs	$198.18	0.06 m	$1,400	0.14 a	**1312.9%**
api/ports/repositories/price_repository_port.py	34	1	**48.31**	9.07 hs	$181.49	0.06 m	$1,400	0.13 a	**1442.8%**
api/services/shopping/extractor_registry.py	29	8	**49.69**	7.34 hs	$146.77	0.05 m	$1,400	0.10 a	**1807.7%**
api/routers/ocr.py	27	5	**51.14**	6.44 hs	$128.83	0.04 m	$1,400	0.09 a	**2073.4%**
api/infrastructure/repositories/sqlalchemy_acc…	26	1	**52.23**	5.92 hs	$118.39	0.04 m	$1,400	0.08 a	**2265.0%**

Métricas Detalladas por Archivo (3/4)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
api/routers/budgets.py	26	1	**52.23**	5.92 hs	$118.39	0.04 m	$1,400	0.08 a	**2265.0%**
api/infrastructure/repositories/sqlalchemy_bu…	23	1	**54.02**	4.82 hs	$96.49	0.03 m	$1,400	0.07 a	**2802.0%**
api/application/use_cases/auth_use_cases.py	22	1	**54.67**	4.47 hs	$89.43	0.03 m	$1,400	0.06 a	**3030.9%**
api/ports/repositories/goal_repository_port.py	21	1	**55.35**	4.13 hs	$82.51	0.03 m	$1,400	0.06 a	**3293.6%**
api/ports/repositories/transaction_repository…	21	1	**55.35**	4.13 hs	$82.51	0.03 m	$1,400	0.06 a	**3293.6%**
api/ports/repositories/wallet_repository_port…	17	1	**58.44**	2.81 hs	$56.29	0.02 m	$1,400	0.04 a	**4874.3%**
api/ports/repositories/account_repository_por…	15	1	**60.27**	2.21 hs	$44.18	0.01 m	$1,400	0.03 a	**6238.0%**
api/domain/entities/user.py	14	1	**61.28**	1.92 hs	$38.41	0.01 m	$1,400	0.03 a	**7190.1%**
api/ports/repositories/budget_repository_port…	13	1	**62.37**	1.64 hs	$32.85	0.01 m	$1,400	0.02 a	**8424.2%**
api/ports/repositories/user_repository_port.py	12	1	**63.54**	1.38 hs	$27.51	0.01 m	$1,400	0.02 a	**10077.2%**
api/domain/exceptions.py	10	1	**66.20**	0.88 hs	$17.60	0.01 m	$1,400	0.01 a	**15812.7%**
api/run_ingesta.py	10	1	**66.20**	0.88 hs	$17.60	0.01 m	$1,400	0.01 a	**15812.7%**
api/services/shopping/__init__.py	10	1	**66.20**	0.88 hs	$17.60	0.01 m	$1,400	0.01 a	**15812.7%**
api/services/sepa/config.py	9	1	**67.74**	0.65 hs	$13.06	0.00 m	$1,400	0.01 a	**21333.3%**
api/ports/extractors/product_extractor_port.py	6	1	**73.67**	0.08 hs	$1.60	0.00 m	$1,400	0.00 a	**175362.8%**
api/domain/entities/__init__.py	3	1	**83.80**	0.00 hs	$0.00	0.00 m	$1,400	0.00 a	**100.0%**
api/wallet_adapters/__init__.py	5	1	**76.34**	0.00 hs	$0.00	0.00 m	$1,400	0.00 a	**100.0%**
js/accounts.ts	1268	462	**0.00**	951.00 hs	$19,020.00	5.94 m	$1,400	13.59 a	**-85.3%**
js/shopping.ts	911	265	**0.00**	683.25 hs	$13,665.00	4.27 m	$1,400	9.76 a	**-79.5%**
js/services/scannerService.ts	827	312	**0.00**	620.25 hs	$12,405.00	3.88 m	$1,400	8.86 a	**-77.4%**
js/goals.ts	755	200	**0.00**	566.25 hs	$11,325.00	3.54 m	$1,400	8.09 a	**-75.3%**
js/budgets.ts	695	173	**0.00**	521.25 hs	$10,425.00	3.26 m	$1,400	7.45 a	**-73.1%**
js/supermercados.ts	675	170	**0.00**	506.25 hs	$10,125.00	3.16 m	$1,400	7.23 a	**-72.3%**
js/reports.ts	468	74	**0.16**	350.27 hs	$7,005.31	2.19 m	$1,400	5.00 a	**-60.0%**
js/transactions.ts	413	170	**0.00**	309.75 hs	$6,195.00	1.94 m	$1,400	4.42 a	**-54.8%**
js/services/shoppingService.ts	350	92	**1.98**	255.56 hs	$5,111.16	1.60 m	$1,400	3.65 a	**-45.2%**
js/shoppingList.ts	324	53	**8.36**	215.92 hs	$4,318.43	1.35 m	$1,400	3.08 a	**-35.2%**
js/scanner.ts	310	75	**6.04**	213.76 hs	$4,275.26	1.34 m	$1,400	3.05 a	**-34.5%**

Métricas Detalladas por Archivo (4/4)

Todas las 7 Métricas Oficiales

Archivo	LOC	G	1. MI	2. Deuda	3. Costo ($)	4. Meses	5. Interés	6. Payback	7. ROI
js/import.ts	302	62	**8.18**	201.81 hs	$4,036.23	1.26 m	$1,400	2.88 a	**-30.6%**
js/services/accountService.ts	244	84	**8.33**	162.67 hs	$3,253.31	1.02 m	$1,400	2.32 a	**-13.9%**
js/ui.ts	221	58	**13.28**	136.40 hs	$2,728.10	0.85 m	$1,400	1.95 a	**2.6%**
js/profile.ts	201	75	**12.38**	125.87 hs	$2,517.38	0.79 m	$1,400	1.80 a	**11.2%**
js/services/shoppingListService.ts	214	32	**17.25**	123.59 hs	$2,471.87	0.77 m	$1,400	1.77 a	**13.3%**
js/ai.ts	206	50	**15.38**	122.81 hs	$2,456.26	0.77 m	$1,400	1.75 a	**14.0%**
js/services/goalService.ts	170	42	**19.27**	94.75 hs	$1,894.95	0.59 m	$1,400	1.35 a	**47.8%**
js/app.ts	157	55	**18.68**	88.42 hs	$1,768.43	0.55 m	$1,400	1.26 a	**58.3%**
js/auth.ts	158	30	**21.95**	83.82 hs	$1,676.37	0.52 m	$1,400	1.20 a	**67.0%**
js/services/authService.ts	128	35	**24.36**	64.82 hs	$1,296.48	0.41 m	$1,400	0.93 a	**116.0%**
js/components/chatbot.ts	102	14	**30.50**	45.39 hs	$907.79	0.28 m	$1,400	0.65 a	**208.4%**
js/utils/utils.ts	83	29	**31.50**	36.11 hs	$722.16	0.23 m	$1,400	0.52 a	**287.7%**
js/services/transactionService.ts	81	26	**32.26**	34.62 hs	$692.44	0.22 m	$1,400	0.49 a	**304.4%**
js/services/sepaService.ts	81	24	**32.53**	34.40 hs	$688.09	0.22 m	$1,400	0.49 a	**306.9%**
js/store/store.ts	68	13	**36.56**	26.14 hs	$522.75	0.16 m	$1,400	0.37 a	**435.6%**
js/services/budgetService.ts	54	7	**40.74**	18.50 hs	$370.01	0.12 m	$1,400	0.26 a	**656.7%**
js/api/apiClient.ts	39	14	**44.56**	11.87 hs	$237.46	0.07 m	$1,400	0.17 a	**1079.1%**
js/api/authApi.ts	40	1	**45.93**	11.63 hs	$232.52	0.07 m	$1,400	0.17 a	**1104.2%**
js/services/cameraService.ts	39	8	**45.36**	11.56 hs	$231.17	0.07 m	$1,400	0.17 a	**1111.2%**
js/services/chatService.ts	30	7	**49.33**	7.70 hs	$154.00	0.05 m	$1,400	0.11 a	**1718.2%**
js/types.ts	1	1	**99.87**	0.00 hs	$0.00	0.00 m	$1,400	0.00 a	**100.0%**